Topic 2 — Mechanics

2.1 Scalars and Vectors

Scalar: A quantity that has magnitude (size) only. Examples: mass, speed, energy, temperature, distance, time, power.
Vector: A quantity that has both magnitude and direction. Examples: force, velocity, acceleration, displacement, momentum, weight.

Adding Vectors

Vectors are added using the **triangle of vectors** (head-to-tail method) or **parallelogram rule** (tail-to-tail method). For two vectors at right angles, use Pythagoras and trigonometry:

$$\text{Resultant} = \sqrt{(A^2 + B^2)} \quad \theta = \tan^{-1}(\text{opposite} / \text{adjacent})$$

For vectors not at right angles, resolve into components first, then add components separately.

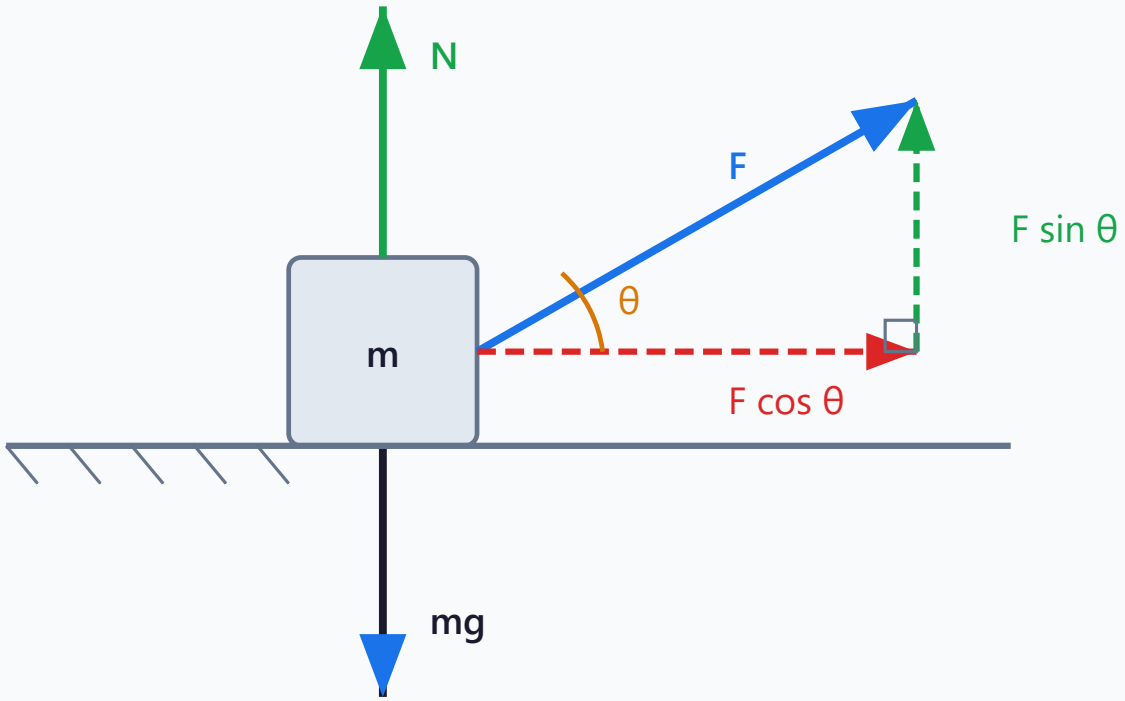
Resolving Vectors into Components

Any vector can be split into two perpendicular components (typically horizontal and vertical):

$$\text{Horizontal: } F_x = F \cos \theta \quad \text{Vertical: } F_y = F \sin \theta$$

where θ is the angle between the vector and the horizontal.

Free Body Diagram: Resolving a Force



Worked Example: A force of 50 N acts at 30 degrees above the horizontal on an object. Find the horizontal and vertical components.

Solution:

$F_x = 50 \cos 30 = 50 \times 0.866 =$ **43.3 N** (horizontal)

$F_y = 50 \sin 30 = 50 \times 0.500 =$ **25.0 N** (vertical)

Check: $\sqrt{(43.3^2 + 25.0^2)} = \sqrt{(1875 + 625)} = \sqrt{2500} = 50$ N. Correct.

Exam Tip: When resolving forces on an inclined plane, resolve parallel and perpendicular to the slope — not horizontally and vertically. The component of weight along the slope is mg sin θ and perpendicular to the slope is mg cos θ (where θ is the angle of the incline).

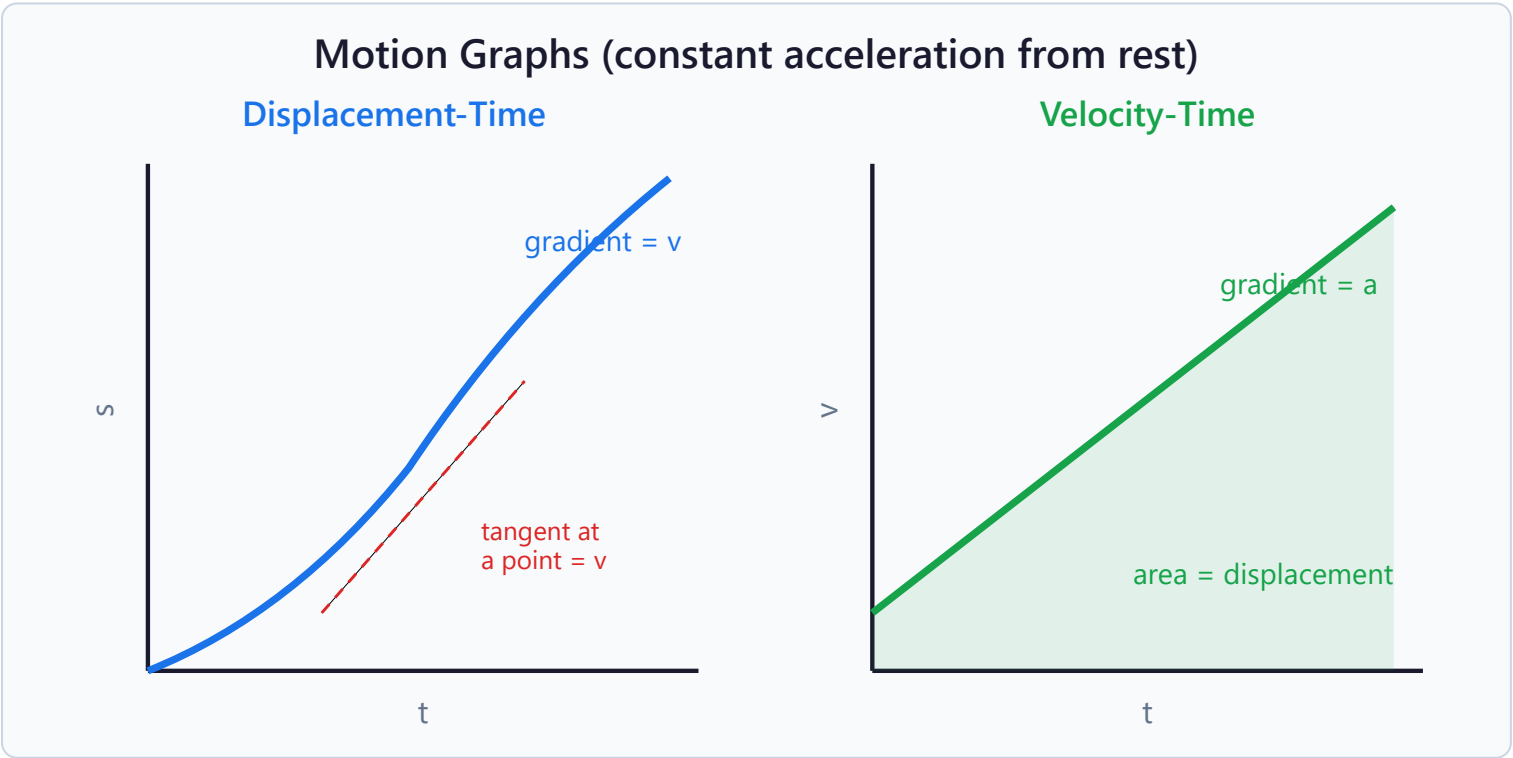
2.2 Motion Graphs

Displacement-Time (s-t) Graphs

- **Gradient = velocity** (instantaneous velocity from the tangent at a point)
- Horizontal line = stationary (v = 0)
- Straight line with constant gradient = constant velocity
- Curve with increasing gradient = increasing velocity (acceleration)
- Negative gradient = moving in the negative direction

Velocity-Time (v-t) Graphs

- **Gradient = acceleration**
- **Area under the graph = displacement** (counting areas below the time axis as negative)
- Horizontal line = constant velocity (a = 0)
- Straight line with positive gradient = constant positive acceleration
- Negative velocity = moving in the opposite (negative) direction



Exam Tip: The area under a v-t graph gives displacement, not distance. If velocity goes negative, the area below the t-axis represents displacement in the negative direction. For distance, take the absolute value of each area section and add them.

2.3 SUVAT Equations (Constant Acceleration)

The SUVAT equations apply to objects moving with **constant (uniform) acceleration** in a straight line. The five kinematic variables are: **s** (displacement), **u** (initial velocity), **v** (final velocity), **a** (acceleration), **t** (time).

Equation	Variables Present	Missing Variable
$v = u + at$	v, u, a, t	s
$s = ut + \frac{1}{2}at^2$	s, u, a, t	v
$s = vt - \frac{1}{2}at^2$	s, v, a, t	u
$v^2 = u^2 + 2as$	v, u, a, s	t
$s = \frac{1}{2}(u + v)t$	s, u, v, t	a

How to Derive Them

- **v = u + at** — from the definition of acceleration: a = (v − u)/t, rearranged.
- **s = ut + ½at²** — area under the v-t graph: s = ut + ½(v − u)t, then substitute v − u = at.
- **v² = u² + 2as** — eliminate t: from v = u + at, get t = (v − u)/a; substitute into s = ½(u + v)t.

Worked Example: A car accelerates uniformly from 5.0 m s^{-1} to 25 m s^{-1} over a distance of 200 m. Calculate (a) the acceleration, (b) the time taken.

Solution:

Known: u = 5.0, v = 25, s = 200. Find a and t.

(a) $v^2 = u^2 + 2as$

$25^2 = 5.0^2 + 2a(200)$

$625 = 25 + 400a$

$a = 600/400 =$ **$1.5\ m\ s^{-2}$**

(b) $v = u + at$

$25 = 5.0 + 1.5t$

$t = 20/1.5 =$ **13.3 s**

Exam Tip: Always list the known values of s, u, v, a, t before choosing which equation to use. Pick the equation that contains the three known quantities and the one unknown. Remember: these equations only work for constant acceleration. If acceleration varies, use calculus or graphical methods.

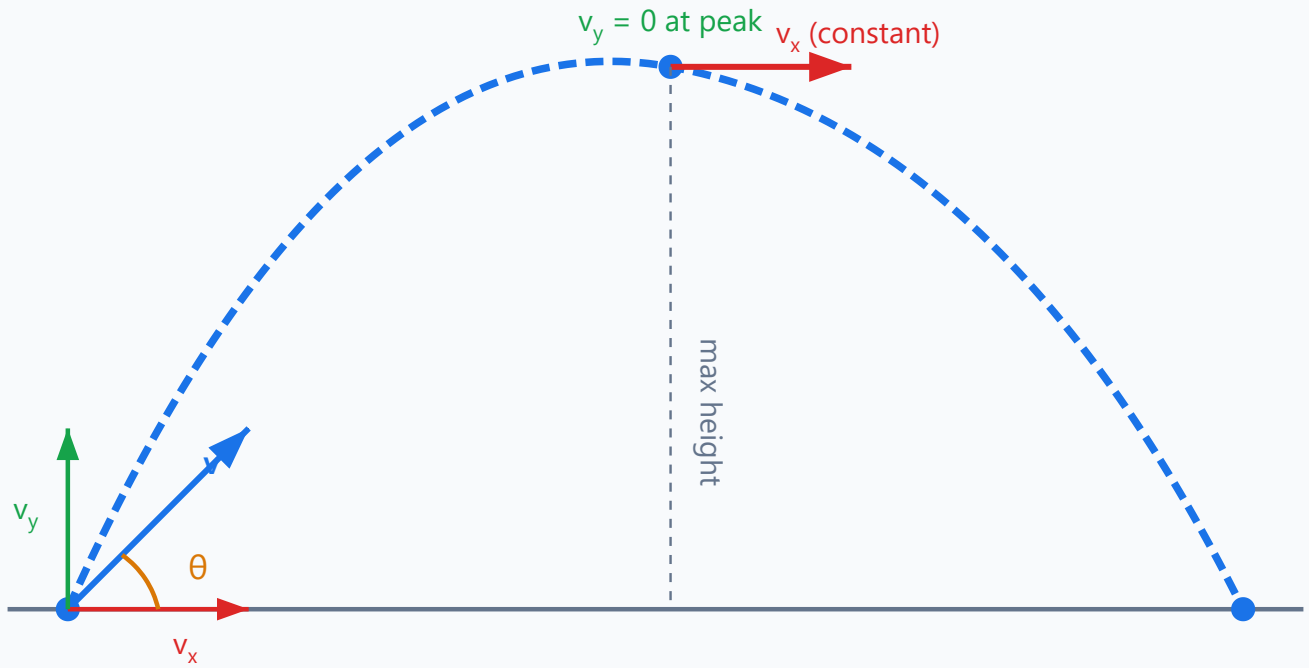
2.4 Projectile Motion

Projectile motion is the motion of an object under the influence of gravity alone (ignoring air resistance). The key principle: horizontal and vertical components of motion are **independent** of each other.

Key Principles

- **Horizontally:** No acceleration (ignoring air resistance), so horizontal velocity is constant: $x = v_x t$
- **Vertically:** Acceleration = g = 9.81 m s^{-2} downwards. Use SUVAT with a = g (or −g depending on sign convention).
- At the highest point, $v_y = 0$ (but v_x remains unchanged throughout).
- Time of flight is determined entirely by the vertical motion.
- The trajectory is a parabola (when air resistance is negligible).

Projectile Motion with Velocity Vectors



Horizontal: constant velocity | Vertical: accelerates at g downward

Projectile Launched at Angle θ to the Horizontal

$$u_x = u \cos \theta \quad u_y = u \sin \theta$$

Then apply SUVAT separately to each component. The horizontal and vertical motions share the same time t.

Worked Example: A ball is kicked at 20 m s^{-1} at 45 degrees to the horizontal from level ground. Find (a) the time of flight, (b) the maximum height, (c) the range. (g = 9.81 m s^{-2})

Solution:

u_x = 20 cos 45 = 14.14 m s^{-1}; u_y = 20 sin 45 = 14.14 m s^{-1}

(a) Time of flight: At landing, vertical displacement = 0. $s_y = u_y t - ½gt^2 = 0$

$t(u_y - ½gt) = 0$, so t = 0 (launch) or $t = 2u_y/g$ = 2 × 14.14/9.81 = **2.88 s**

(b) At max height, $v_y = 0$. $v_y^2 = u_y^2 - 2gs$

$0 = 14.14^2 - 2(9.81)s_{max}$

$s_{max} = 200/19.62 =$ **10.2 m**

(c) Range = $u_x \times t = 14.14 \times 2.88 =$ **40.7 m**

Exam Tip: For a horizontal launch (e.g. ball rolling off a table), $u_x = u$ and $u_y = 0$. The time to hit the ground depends only on the height: $h = ½gt^2$. The range is then $R = u_x \times t$. A common mistake is to use the launch speed as the vertical component.

2.5 Newton's Laws of Motion

Newton's First Law: An object remains at rest or moves with constant velocity unless acted upon by a resultant (net) force. This defines inertia — the tendency of an object to resist changes in its motion.

Newton's Second Law: The resultant force on an object equals the rate of change of its momentum. For constant mass: **F = ma**. More generally: F = dp/dt.

Newton's Third Law: If object A exerts a force on object B, then object B exerts an equal and opposite force on object A. These forces are always: the same type, equal in magnitude, opposite in direction, and act on **different objects**.

Free Body Diagrams

A free body diagram shows all the forces acting on a single isolated object:

- **Weight (W = mg)** — acts vertically downwards from the centre of mass
- **Normal contact force (N)** — acts perpendicular to the surface of contact
- **Friction (f)** — acts parallel to the surface, opposing relative motion or tendency to slide
- **Tension (T)** — acts along a string or rope, pulling away from the object
- **Air resistance / drag (D)** — acts opposite to the direction of motion

- **Thrust / driving force (F)** — acts in the direction of motion

Exam Tip: Weight and normal reaction are NOT a Newton's Third Law pair. They act on the same object and are different types of force. A correct N3L pair for weight: "The Earth pulls the object down (gravity), and the object pulls the Earth up (gravity) with an equal force." For the normal force: "The surface pushes the object up, and the object pushes the surface down."

Worked Example: A 1200 kg car accelerates from rest at 2.5 m s^{-2}. The total resistive force (friction + drag) is 400 N. Calculate the driving force produced by the engine.

Solution:
Resultant force = ma = 1200 × 2.5 = 3000 N
Resultant = driving force − resistive force
3000 = F_{drive} − 400
F_{drive} = **3400 N**

2.6 Momentum

Momentum (p): The product of an object's mass and velocity. It is a vector quantity. Units: kg m s^{-1} (or equivalently N s).

$$p = mv$$

Conservation of Momentum

In a **closed system** (no external resultant force), the total momentum before an interaction equals the total momentum after:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Types of Collision

Type	Momentum Conserved?	KE Conserved?	Example
Perfectly elastic	Yes	Yes (no KE lost)	Ideal gas molecule collisions; billiard balls (approximately)
Inelastic	Yes	No (some KE converted to heat, sound, deformation)	Most real collisions; car crashes
Perfectly inelastic	Yes	No (maximum KE lost; objects stick together)	Bullet embedding in a block; coupling railway carriages
Explosion/separation	Yes (total = 0 if initially at rest)	No (KE increases; from internal/chemical energy)	Guns firing; rocket propulsion; spring-loaded trolleys

Worked Example: A 2.0 kg trolley moving at 3.0 m s^{-1} collides with a stationary 1.0 kg trolley. They stick together. Find (a) the velocity after the collision, (b) the KE lost.

Solution:

(a) Conservation of momentum: $m_1u_1 + m_2u_2 = (m_1 + m_2)v$

$2.0 \times 3.0 + 1.0 \times 0 = (2.0 + 1.0)v$

$6.0 = 3.0v$, so $v =$ **2.0 m s^{-1}**

(b) KE before = $\frac{1}{2}(2.0)(3.0)^2 = 9.0$ J

KE after = $\frac{1}{2}(3.0)(2.0)^2 = 6.0$ J

KE lost = 9.0 − 6.0 = **3.0 J** (converted to heat and sound; this was an inelastic collision)

2.7 Impulse

Impulse: The change in momentum of an object. It equals the force multiplied by the time for which it acts. Units: N s (or kg m s^{-1}).

$$\text{Impulse} = F\Delta t = \Delta p = mv - mu$$

The area under a **force-time graph** equals the impulse (change in momentum).

Safety applications: Crumple zones, airbags, helmets, and crash mats all work on the same principle — they increase the time of impact Δt, which reduces the average force F for the same change in momentum (since F = Δp/Δt).

2.8 Work, Energy, and Power

Work Done

Work done (W): The energy transferred when a force moves an object through a distance in the direction of the force. Units: J (joules).

$$W = Fd\cos\theta$$

where θ is the angle between the force and the direction of displacement. If θ = 0, W = Fd. If θ = 90, W = 0 (no work done — e.g. centripetal force does no work).

Kinetic Energy

$$E_k = \tfrac{1}{2}mv^2$$

The work-energy theorem: the net work done on an object equals its change in kinetic energy: $W_{net} = \Delta E_k$.

Gravitational Potential Energy

$$E_p = mgh$$

where h is the vertical height change above a chosen reference point. This is the "near-surface" approximation valid when g is approximately constant.

Conservation of Energy

Energy cannot be created or destroyed, only transferred from one form to another. In the absence of dissipative forces (friction, air resistance):

$$\tfrac{1}{2}mv^2 + mgh = \text{constant}$$

Worked Example: A 0.50 kg ball is dropped from a height of 8.0 m. Assuming no air resistance, find (a) its speed just before hitting the ground, (b) the time it takes to fall.

Solution:

(a) Using conservation of energy: $mgh = \tfrac{1}{2}mv^2$

$v = \sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 8.0)} = \sqrt{(156.96)} =$ **$12.5\ m\ s^{-1}$**

(b) Using $s = \tfrac{1}{2}gt^2$: $8.0 = \tfrac{1}{2}(9.81)t^2$

$t = \sqrt{(16.0/9.81)} = \sqrt{(1.631)} =$ **1.28 s**

Power

Power (P): The rate of doing work (or the rate of energy transfer). Units: W (watts). 1 W = 1 J s^{-1}.

$$P = W/t = \Delta E/\Delta t$$

$$P = Fv$$

The second equation (P = Fv) is very useful for finding the driving force at a given speed, or the power needed to maintain a constant velocity against resistive forces.

Efficiency

$$\text{Efficiency} = (\text{useful energy output} / \text{total energy input}) \times 100\%$$

$$\text{Efficiency} = (\text{useful power output} / \text{total power input}) \times 100\%$$

Efficiency is always less than 100% in practice because some energy is always dissipated (usually as thermal energy due to friction).

Worked Example: A car of mass 1500 kg travels at a constant velocity of 30 m s^{-1} on a level road. The total resistive force is 800 N. Calculate (a) the driving force, (b) the power output of the engine, (c) the rate of fuel consumption if the fuel provides 40 MJ per litre and the engine is 25% efficient.

Solution:

(a) At constant velocity, a = 0, so driving force = resistive force = **800 N**

(b) P = Fv = 800 × 30 = **24000 W** = 24 kW

(c) Total power needed = 24000 / 0.25 = 96000 W (since only 25% is useful)

Fuel energy per second = 96000 J s^{-1}

Volume per second = 96000 / (40 × 10^6) = 2.4 × 10^{-3} L s^{-1}

= **8.64 L per hour** (or about 0.144 L per minute)

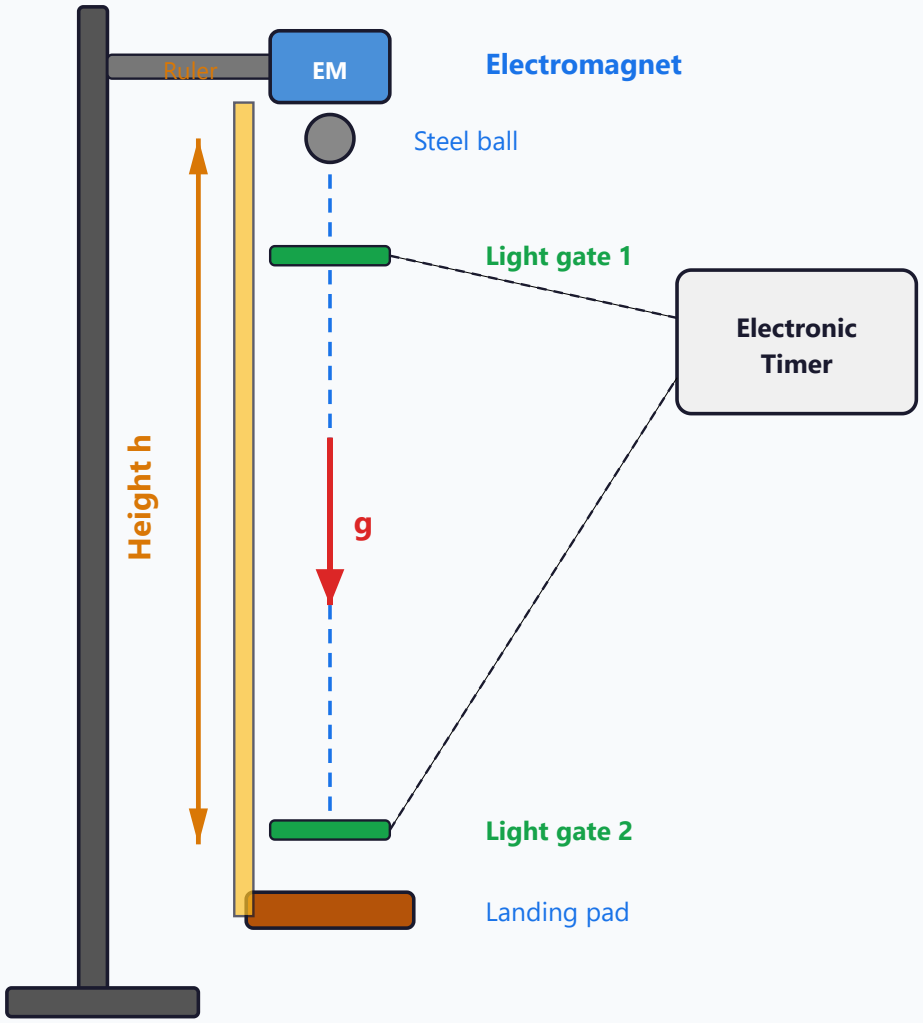
Exam Tip: At constant velocity, the resultant force is zero, so driving force = resistive force. Power at constant velocity is P = Fv where F is the driving force. If the car accelerates, the driving force must exceed the resistive force, and P = Fv gives the instantaneous power at speed v.

Summary of Key Equations

Quantity	Equation	Units
SUVAT	$v = u + at$; $s = ut + ½at^2$; $v^2 = u^2 + 2as$; $s = ½(u+v)t$	m, m/s, m/s^2, s
Newton's Second Law	$F = ma = \Delta p/\Delta t$	N
Momentum	$p = mv$	kg m s^{-1}

Impulse	$F\Delta t = \Delta p$	N s
Work done	$W = Fd\cos\theta$	J
Kinetic energy	$E_k = ½mv^2$	J
GPE (near surface)	$E_p = mgh$	J
Power	$P = W/t = Fv$	W
Efficiency	η = useful output / total input	% or ratio

⚗ Core Practical 1: Determine the acceleration of a freely falling object



CP1: Measuring g by free fall

Aim

To determine the acceleration due to gravity (g) by timing the free fall of a ball bearing through a known height.

Apparatus

Electromagnet and steel ball bearing, two light gates (or electromagnet + trapdoor timer), electronic timer or datalogger, metre ruler, clamp stand, boss and clamp, set square, landing pad.

Method

1. Clamp the electromagnet at the top. Hold the ball bearing with it.

2. Position two light gates at a known vertical separation h, measured with a metre ruler and set square.
3. Connect both light gates to the timer (measures the time interval between the ball passing each gate).
4. Switch off the electromagnet to release the ball. Record the fall time t.
5. Repeat 3 times for each height and calculate the mean time.
6. Vary the height h (at least 5–6 different values from ~0.20 m to ~1.00 m).

Analysis

Using $s = ut + ½gt^2$ with $u = 0$: $h = ½gt^2$, so $2h = gt^2$.

Plot **2h (y-axis) vs t^2 (x-axis)**. Gradient = g. Expected: ~9.81 m s^{-2}.

Safety

- Cushion beneath landing area. Stable clamp stand (G-clamp). Feet clear of falling ball.

Sources of Error & Improvements

Source of Error	Type	Improvement
Residual magnetism delays ball release	Systematic (g too small)	Use two light gates (eliminates release timing); or use non-magnetic release
Air resistance on ball	Systematic (g too small)	Use a small, dense, smooth steel ball
Height measurement to wrong point	Random	Use a set square; measure from centre of each light gate
Ball not falling vertically through both gates	Systematic	Use a plumb line to align the gates vertically

Exam Tip: If the 2h vs t^2 graph has a non-zero y-intercept, this indicates a systematic timing error (e.g. release delay). The gradient is still valid for determining g. Compare your result

with 9.81 m s^{-2} and calculate the percentage difference.